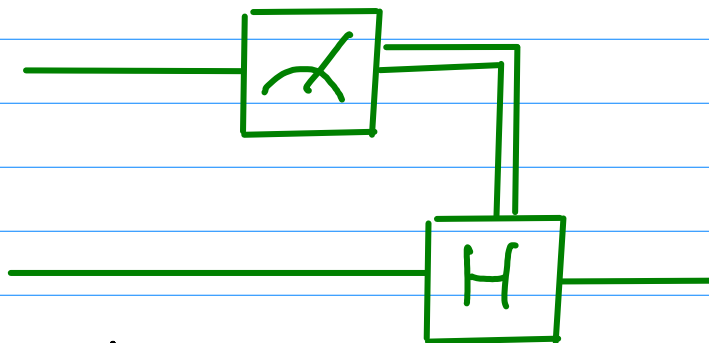
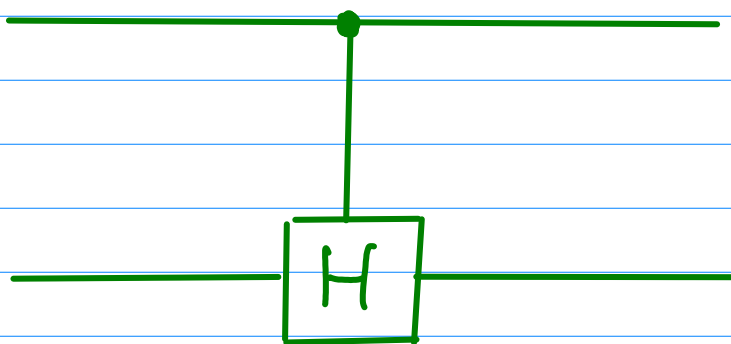


2 A possible Clifford circuit is



If delayed measurement were possible, this would imply we had the gate



$$= |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes H = U$$

$$U = I \otimes I + \sigma_z \otimes I + I \otimes H - \sigma_z \otimes H \\ = I \otimes |H\rangle\langle H| + \sigma_z \otimes |H\rangle\langle H|$$

$$\text{Using } H = \frac{1}{\sqrt{2}}(|H\rangle + |X\rangle) - \frac{1}{\sqrt{2}}(|H\rangle - |X\rangle)$$

$$U \sigma_x \otimes I U^\dagger = \sigma_x \otimes |H\rangle\langle H| - \sigma_x \otimes |H\rangle\langle H| \\ = \sigma_x \otimes I$$

$\therefore U$ is not Clifford